ModelIt!

Student Activity Pack

Renewable Energy Grid Optimization: Powering the Future

Modeling the Complex Challenge of a 100 Percent Clean Energy Grid

Grade 09L2 | NGSS HS-PS3-3, HS-ETS1-3 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why can't we just replace all power plants with solar panels?

2 What happens to electricity supply when the sun sets or the wind stops?

3 Draw a diagram showing how electricity gets from a solar panel to your home.

4 What are the pros and cons of different energy sources?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Solar Output | Wind Output | Battery Storage | Grid Demand | Fossil Fuel Backup | Transmission Efficiency | Cost per kWh | Carbon Emissions

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Sunny Windy Day

Peak solar, strong wind, moderate demand

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Worst Case

Zero solar, calm wind, peak demand

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Optimization

Vary mix to minimize cost and emissions

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Hydrogen Storage — Long-duration energy storage using excess renewable electricity to produce hydrogen via electrolysis

Nuclear Baseload — Constant-output nuclear power providing reliable baseload electricity independent of weather

Demand Response — Smart grid systems that shift flexible electricity demand to match renewable availability

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

Why can't we just replace all fossil fuel power plants with solar panels and wind turbines tomorrow? Use evidence from your simulation to support your answers.

1. **Why is the last 20 percent of decarbonization so much harder than the first 80 percent?**

2. **What role does energy storage play in enabling higher renewable penetration?**

3. **How do trade-offs between cost, emissions, and reliability constrain grid design?**

4. **Why might geographic diversity of renewable installations reduce intermittency risk?**

5. **What technology breakthroughs would most change your optimal grid design?**

8 STEM Challenge: Design a Regional Clean Energy Grid

ModelIt!

YOUR ENGINEERING MISSION

Design the optimal energy mix for a region that achieves maximum renewable penetration while maintaining grid reliability and minimizing cost.

THE SCENARIO

A region of 5 million people must achieve 90 percent clean energy by 2035. Design the energy portfolio: how much solar, wind, storage, and backup do you need? What does it cost? How reliable is it during extreme weather?

Engineering Design Thinking

1. HOW WOULD YOU DISTRIBUTE SOLAR AND WIND INSTALLATIONS GEOGRAPHICALLY?
2. WHAT TYPES OF STORAGE WOULD YOU DEPLOY -- BATTERIES, PUMPED HYDRO, HYDROGEN?
3. HOW WOULD YOU HANDLE SEASONAL VARIATION IN SOLAR OUTPUT?
4. WHAT IS THE MINIMUM FOSSIL FUEL BACKUP NEEDED TO MAINTAIN 99.97 PERCENT RELIABILITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Grid Engineers and Energy Systems Analysts design and optimize electrical grids for the clean energy transition, earning $75,000-$140,000/year at utilities, grid operators, energy companies, and government agencies.